Roll No. | | | | | | | | | | | Total No. of Pages: 02

Total No. of Questions : 13

B.Pharmacy (Sem.–3)

PHARMACEUTICAL MICROBIOLOGY

Subject Code : BP-303T

M.Code : 75107

Date of Examination:20-12-2024

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Answer briefly:**

   a) Differentiate between prokaryotes and eukaryotes.

   b) Explain the ultra structure of Bacteria.

   c) What is the importance of laminar air flow?

   d) MIC

   e) Sterlization

   f) Grams staining

   g) Disinfectants

   h) Antimicrobial agents

   i) Suggest a suitable method for the sterility testing of ophthalmic and solids.

   j) Suggest a suitable method for the sterilization of 0.9% w/v sodium chloride injection and petriplates.

(S29)-1915

SECTION-B

2. a. Explain the application of cell cultures in pharmaceutical industry and research.

   b. Enumerate any one equipments for the large scale sterilization of pharmaceuticals.

3. Write short note on

   a. Sterilization by filtration.

   b. Microbial Assay of vitamins.

4. Enumerate the methods for the standardization of antibiotics.

SECTION-C

5. Distinguish between :

   a. Toxins and toxoid

   b. Sterility testing and sterilization

6. Explain the bacterial growth curve. What are the methods to identify bacteria?

7. Explain the different methods for the evaluation of disinfectants.

8. Highlight principle, construction, working and validation of horizontal laminar air flow.

9. What are the factors affecting the microbial spoilage of pharmaceutical products?

10. Explain the assessment of microbial contamination and spoilage.

11. Explain the sterility testing according to IP standards.

12. Enumerate morphology, classification and cultivation of fungi.

13. Write short note on sterility indicators.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

College of Pharmacy LIBRARY

DEC-2024